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Date

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1). Menghitung mean populasi (μ) dan Varians Populasi (σ^2)

- Data 98, 99, 97, 98, 99, 99, 97, 97, 98, 99
- $N = 10$

$$\mu = \frac{\sum x}{N} = \frac{98 + 99 + 97 + 98 + 99 + 98 + 97 + 97 + 98 + 99}{10}$$

$$= \frac{980}{10}$$

$$= 98$$

- Deviasi kuadrat $(x - \mu)^2$:
  - $= (97 - 98)^2 = 1$ (sebanyak 3 data) $\rightarrow 3 \times 1 = 3$
  - $= (98 - 98)^2 = 0$ (sebanyak 4 data) $\rightarrow 4 \times 0 = 0$
  - $= (99 - 98)^2 = 1$ (sebanyak 3 data) $\rightarrow 3 \times 1 = 3$
- Total $\sum (x - \mu)^2 = 3 + 0 + 3 = 6$
- $\sigma^2 = \frac{\sum (x - \mu)^2}{N} = \frac{6}{10} = 0,6$
- $\sigma = \sqrt{0,6} \approx 0,7746$
- Mean dari Sampel
  - $\mu_{\bar{x}} = \mu = 98$
- Standar deviasi ($\sigma_{\bar{x}}$):

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N - n}{N - 1}}$$

- Total akhir

$$\sigma_{\bar{x}} \approx 0,5164$$

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2). Diket :

$\mu = 2.000$, $\sigma = 600$, $n = 100$

1. Standar deviasi sampel

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{600}{\sqrt{100}} = 60$$

2. Nilai z

- $z_1 = \frac{1.900 - 2.000}{60} = -1,67$

- $z_2 = \frac{2.050 - 2.000}{60} = 0,83$

3. Probabilitas

$P(-1,67 < z < 0,83) = P(z < 0,83) - P(z < -1,67)$

$= 0,7967 - 0,0475 = 0,7492$

$= 0,7492 \,/\, 74,92\%$

3). Diket :

$n_1 = 140$, $\bar{x}_1 = 163$, $\sigma_1 = 5,2$

$n_2 = 140$, $\bar{x}_2 = 152$, $\sigma_2 = 4,9$

1. Beda mean :

$\mu_1 - \mu_2 = 163 - 152 = 11$

2. Standar deviasi beda mean

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{5,2^2}{140} + \frac{4,92}{140}} = \sqrt{\frac{27,04 + 24,01}{140}} =$$

$\sqrt{0,3696} \approx 0,6038$

3. Nilai z (d = 10) :

$$z = \frac{10 - 11}{0,6038} = -1,66$$

4. $P(z \geq -1,66 = 1 - 0,048 = 0,9515$

$0,9515 \,/\, 95,15\%$

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4) - Diket :

$P_1 = 0.35$, $P_2 = 0,30$, $n_1 = 250$

1. beda proporsi populasi

$P_1 - P_2 = 0,35 - 0,30 = 0,05$

2. Standar deviasi beda proporsi

$$\sigma_{P_1 - P_2} = \sqrt{\frac{0,35(0,65}{200} + \frac{0,30(0,70}{270}} = \sqrt{\frac{0,2275 + 0,2100}{250}}$$

$$= \sqrt{0,00175} \approx 0,0418$$

3. nilai Z ($d = 0,02$)

$$Z = \frac{0,02 - 0,05}{0,0418} = -0,72$$

4. Probabilitas

$P(Z < -0,72 = 0,2358$

$= 0,2358 / 23,58\%$

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